Electrocardiogram (ECG)

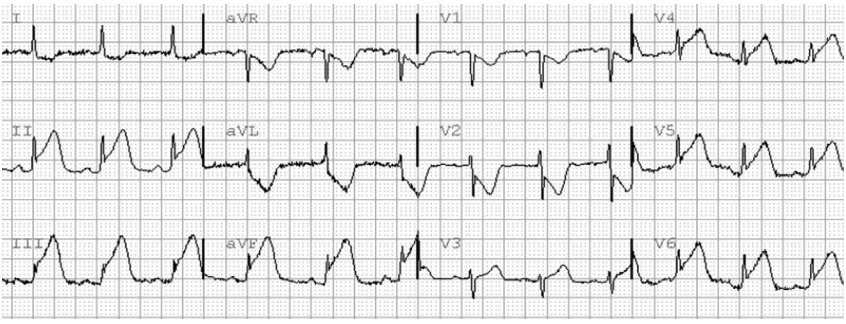
INTRODUCTION

The ECG is a vital part of the initial diagnosis and evaluation of individuals with chest pain. The resting ECG has proven its effectiveness as a diagnostic tool for detecting heart disease as part of the normal routine examination approach in cardiac medicine. In addition to their clinical use, ECGs from apparently healthy subjects have been utilized to explore the prevalence, correlates, and predictive value of asymptomatic heart disorders in the general population.

ACUTE MYOCARDIAL ISCHEMIA AND THE ECG

The P wave, the PR interval, the QRS complex, the ST segment, and the T and U waves are all parts of the electrical activation of the heart, may be altered by acute myocardial ischemia. The most dramatic ECG symptoms of acute transmural myocardial ischemia are ST-segment elevation (STE) in the leads facing the ischemic zone and ST-segment depression (STD) in the leads facing the anatomically opposite myocardial segments.. Subendocardial ischemia, on the other hand, can cause STD in the ECG leads nearest to the affected zone. The ECG may also be normal or reveal very slight changes, especially if the patient is asymptomatic at the time the ECG is taken. This, together with other evidence, has led to the conclusion that STE on the whole is a good thing. STE on the presenting ECG in patients with compatible symptoms appears to indicate "ongoing transmural myocardial ischemia," necessitating immediate reperfusion therapy, whereas patients without STE can be treated more cautiously, at least initially, unless they are hemodynamically unstable.. As a reason, the ECG has become increasingly important in the initial assessment and triage of patients with symptoms consistent with chest discomfort, in addition to a thorough history and physical examination. An ECG should be done and interpreted by an experienced physician within 10 minutes after arriving at the Emergency Department, according to current guidelines. Furthermore, prehospital ECG transmission (or interpretation) is suggested to shorten the period between first medical contact and reperfusion therapy (catheterization laboratory activation before surgery).

However, it's important to keep in mind that "ongoing transmural myocardial ischemia," that usually manifests as STE, should be identified and treated instead of the ST segment itself. Many people have STE at baseline even if they don't have acute myocardial ischemia. Other than acute myocardial ischemia, dynamic ST alterations can occur. Furthermore, those with nonischemic STE at baseline may experience chest discomfort; some will be labelled with noncardiac pain, while others will develop non-STE acute coronary syndrome (NSTE-ACS) or STE-ACS. Based on epidemiological data, the recommendations established STE thresholds. Because a high proportion of healthy people (particularly young males) show mild STE in leads V2–V3, a threshold of 0.25 mV in men under 40 years, and 0.2 mV for women under 40 years is advised.



STE above the threshold in at least two adjacent ECG leads should raise suspicion of STE-ACS. However, it should be noted that a large proportion of patients may have STE over this threshold without developing acute STE-ACS, whereas patients with STE less than the accepted thresholds may have acute continuing transmural ischemia due to a abrupt closure of an epicardial coronary artery.. Small QRS complexes in the affected leads, especially in inferior STE-ACS, may suggest myocardial protection due to preconditioning or residual collateral circulation, among other explanations. While new or probably new LBBB has been regarded a substitute for ST-segment elevation myocardial infarction (STEMI), most occurrences of LBBB at the time of presentation are "unknown to us."

LBBB at presentation is usually caused by acute epicardial coronary blockage, can interfere with STE analysis, and should not be used alone to diagnose STEMI-equivalent, according to recent guidelines. The differential diagnoses of nonischemic STE were extensive. In many cases, an experienced electrocardiographer can easily recognise typical ECG patterns, such as prominent J-points with STE in the lateral leads (an early repolarization pattern) or diffuse STE with STD in leads aVR and V1 with PR segment depression (an acute pericarditis pattern). However, in many other cases, there could be multiple causes for the STE, making a definitive diagnosis impossible at the time of presentation. Hypertension, for example, is a very well condition.

Patients with ECG symptoms of left ventricular hypertrophy, on the other hand, may have an anterior STEMI. It is commonly assumed that comparing the current ECG tracing to earlier ECG tracing will allow the correct diagnosis to be made. Nonischemic STE, on the other hand, often exhibits dynamic changes.. STE due to early repolarization, for example, may not be visible at a high heart rate or after hyperventilation, but STE caused by a ventricular aneurysm would, and so on. To determine if acute reperfusion therapy should be used in these cases, the clinician may rely on the child's history and physical examination or order a transthoracic echocardiography. Although it has been reported that the percentage of patients with incorrect activation of the p-PCI protocol due to nonischemic STE ranges from 14 percent to over 40%

Even with significant stenosis, not all patients with positive cardiac markers who present with STE and are found to have angiographically verified coronary artery disease have true STE-ACS. Some people with nonischemic STE may have NSTE-ACS, as a baseline, a condition known as "pseudo-STEMI." The pattern of ST- segment deviation in adjudicated true STE-ACS should be compatible with the site of coronary artery occlusion, and ECG abnormalities should resolve if early reperfusion therapy is successful. Even transmural ischemia owing to full blockage of an epicardial artery may resolve without (substantial) increase of cardiac markers if the coronary artery occlusion recanalizes swiftly, either spontaneously or with reperfusion therapy. 12 STE can be accompanied by stress-induced (takotsubo) cardiomyopathy.

ST-ELEVATION ACUTE CORONARY SYNDROME

The QRS complex, the ST-segment, and the T-wave shape may all be affected by acute transmural ischemia.. Based on the pattern of ECG alterations, the reader can determine the location of a coronary artery obstruction, the size and extent of the ischemic area at risk, the severity of ischemia, and the prognosis.

Changes in the Initial Portion of the QRS

Q waves are believed to be a sign of permanent necrosis in the chronic phase of myocardial infarction. About half of patients who present within one hour of the onset of STE-ACS show Q waves in the leads, especially in the anterior leads. These Q waves could be transient, and therefore don't always mean irreparable harm.. Intense ischemia is thought to cause a brief lack of electrical activity in the at-risk zone ("myocardial concussion"). As a result, Q waves on presentation could indicate irreparable damage and/or a significant ischemic zone, indicating a massive ultimate infarction. Preexisting Q waves in inferior STE-ACS, on the other hand, may vanish during acute ischemia—the Q waves may be "dragged up" by the injury current—and preexisting Q waves in superior STE-ACS may vanish during acute ischemia.

Changes in the Terminal Portion of the QRS

The terminal component of the QRS is In leads with terminal S waves, this manifests as a reduction or elimination of the S wave, and In leads with terminal R-wave configuration, a J/R wave ratio of 50% or more is needed. Despite successful epicardial coronary artery recanalization, patients with significant terminal QRS distortion (Sclarovsky–Birnbaum grade III ischemia) have more severe ischemia and less myocardial salvage., and a worse prognosis than those with STE-ACS without terminal QRS distortion (Sclarovsky–Birnbaum grade II).

ST Segment Deviation

The most well-known symptom of acute transmural ischemia is STE in leads confronting the ischemic zone. Ischemia caused by blockage of the left anterior descending (LAD) coronary artery causes STE in the precordial leads, particularly in leads V2–V4. In anterior STEMI, synchronous STE in leads I and aVL, as well as STD in the inferior leads (II, III, and aVF), suggests LAD blockage proximal to the first diagonal branch. The sensitivity of this indication, however, is limited because, in many cases with a long, looping LAD, simultaneous inferior wall ischemia tends to oppose the signals.. In anterior STEMI, simultaneous STE in leads I and aVL, as well as STD in the inferior leads (II, III, and aVF), suggests LAD blockage proximal to the first diagonal branch. However, because simultaneous inferior wall ischemia tends to oppose and cancel ST deviation in leads I and aVL in many cases with a long, looping LAD, the sensitivity of this indication is poor. The dominant left circumflex (LCx) and right coronary arteries (RCA) are often blocked. STE in lead V1, STD attenuation in leads V2–V3, and STE in the right-sided precordial leads V3R and V4R can all be indications of right ventricular involvement caused by proximal RCA blockage.There may be minimal or no STE in the inferior leads in relatively rare cases of AMI limited to the lateral zone, and the ECG presentation of ischemia will be reciprocal STD in leads V1–V3. 11 Leads V7–V9 (posterior chest leads) may detect STE in many cases. Patients with STD in leads V1–V3 4 or V1–V4 3 should be deemed STE-ACS equivalents and referred for emergency revascularization, according to the guidelines. However, there is still some ambiguity about this pattern. It's unclear whether true lateral STE-ACS equivalent can be considered out in circumstances when the STD extends beyond lead V4 (V5–V6). STD in leads V1–V3 is also common in patients with right bundle branch block (RBBB). In individuals with complete or incomplete RBBB, there are no criteria for diagnosing or excluding acute lateral STE-ACS equivalent. Although the extent of the ischemia area is frequently associated with risk, this is not always the case. The ECG leads record the vector of the heart's global activation toward and away from the electrode, not local events. The cancellation of ST deviation in leads facing these segments may be caused by concurrent ischemia of opposing segments.

Changes in the T and U Waves

An increase in the amplitude of the T waves is the first ECG manifestation of acute transmural ischemia. 21 In most cases, STE adopts this modification quickly. 15 A tiny minority of patients, however, may not experience any significant ST-segment variation. This pattern is seen in patients who have established rich collateral circulation due to persistent subtotal blockage of the artery. 22 The pattern is difficult to distinguish since the amplitude of the T waves in the various leads varies greatly among normal patients. Furthermore, factors other than ischemia (such as serum potassium) have an impact on T-wave amplitude. Tall peaked T waves can occasionally be identified in a patient with compatible symptoms by comparing past ECGs. STE and inversion of the terminal section of the T waves are seen in certain cases. This pattern's significance is unknown. An ECG sign of reperfusion is an early inversion of the terminal segment of the T waves, as well as a decrease in STE. 14,23 However, it is uncertain whether patients with STE and negative T waves should be sent for emergency reperfusion therapy because the culprit artery has already been recanalized. The guidelines propose immediate reperfusion treatment for all patients with STE All patients with STE who have compatible symptoms that began within 12 hours of presentation (whether or not symptoms have resolved) should receive prompt reperfusion treatment, according the guidelines (the morphology of the T waves is not mentioned) The U has been inverted.

NON-ST ELEVATION ACUTE CORONARY SYNDROME

Ischemia is intermittent in many patients with NSTE-ACS, as said before. Many patients do not have active ischemia at the time of presentation. As a result, ECG changes often are minor or reflect a state of reperfusion in many patients (inversion of the terminal portion of the T waves or nonspecific T changes without significant ST-segment deviation).). ECG alterations, such as ST-segment deviation and changes in T- and U-wave morphology, can be identified during active ischemia. As a result, it's crucial to note whether the ECG was taken while the patient was symptomatic or not. Serial ECG recording during and after the onset of symptoms may reveal minor changes that would otherwise go unnoticed. Acute ischemia of the subendocardium caused by a partial reduction in flow and/or an increase in demand.

Any STD can be represented as STE using this method. However, this does not make the patient have STE-ACS, which is defined as an epicardial coronary artery blockage with continuous transmural ischemia (see above). In patients with isolated STE in the mirror leads, it remains to be shown if emergency reperfusion treatment is useful and cost-effective (isolated STD in the conventional leads). T-wave morphology is also affected by acute subendocardial ischemia. STD

with tall peaked positive T waves is now regarded as a marker of subendocardial ischemia in the region. 15, 25, and 26. In patients with (prominent) hyper-acute T waves, which may precede STE, the European STEMI guidelines advocate repeat ECGs or STE monitoring.

The shape of the T wave has been suggested as a possible tool for distinguishing between anterior subendocardial ischemia (due to subtotal LAD blockage) and lateral transmural ischemia (due to LCx or RCA occlusion) (Fig. 6). Keep in mind that T-wave morphology changes with time and with reperfusion. In the some cases, the T wave is biphasic (either above the isoelectric line initially with terminal negative portion or initially negative with terminal positive deflection). Different researchers utilised different definitions for positive and negative T waves; some focused at the T wave's terminal component, while others focused at the area under the curve to see whether it was positive or negative. Several authors also failed to create a definition for the T wave. Several distinct ECG patterns seen in patients with NSTE-ACS have been characterized:

1. With isoelectric ST segments or slight ST deviation, T-wave inversion occurs. Isolated T-wave inversion is not regarded an ominous symptom in NSTE-ACS. Flat/mild isolated T-wave inversion, especially in leads with significant R waves, is often found after symptoms have subsided and is seen as an indication of reperfusion.

   On angiography, a significant stenosis of the LAD coronary artery is detected in ACS patients who present with deep isolated T-wave inversion in V1 to V3/V4 (the "Wellens' sign"). These instances are classified as NSTE-ACS by definition, yet they are more often than not an evolutionary pattern of STEMI with spontaneous or therapy-induced ST resolution. These patients should not be evaluated invasively right away, according to the guidelines. Reocclusion can occur with the development of positive T waves or even STE in some circumstances, regardless of the fact that this pattern is mostly seen after symptoms have resolved.. As a result, this pattern (deep negative T wave in V1 through V3–V4) could be interpreted as an anterior STE-ACS that "spontaneously reperfused." However, unlike traditional STE-ACS, there is currently no reason for initiating reperfusion therapy within 60 minutes after arrival.
2. Up-sloping STD with positive tall T waves: Up-sloping STD with positive T waves is prevalent in people who are experiencing tachycardia, even if they don't have coronary artery disease, and it's usually not taken as a sign of myocardial ischemia. However, at slower heart rates, the same pattern can be seen in patients with NSTE-ACS, and it's becoming more well recognised as a marker of regional

subendocardial ischemia. 25 This pattern has been linked to subtotal or total obstruction of the LAD, resulting in subendocardial ischemia, when shown in the precordial leads. Patients with LCx lesions, on the other hand, show a similar pattern (Fig. 6). 15 It's important to remember that this pattern might be unstable, and STE can develop if the culprit artery goes from a stenosis to a stenosis. Because patients with ACS may have reactive tachycardia and tachycardia can cause ischemia, distinguishing between a "It's essential to get a "typical" up-sloping STD accompanied tachycardia and localised subendocardial ischemia. However, no heart rate threshold has been established to differentiate these two entities..

3. STD in the inferior and anterolateral leads with ST elevation in lead aVR: During episodes of chest pain, widespread STD in 6 leads is commonly coupled with inverted T waves and STE in lead aVR, indicating pervasive subendocardial ischemia caused by left main, left main equivalent, or severe three vascular disease-related ischemia. When a patient is asymptomatic, the ECG may be normal or show simply T-wave inversion. These patients are at a very high risk of experiencing cardiogenic shock and/or ventricular fibrillation, hence immediate coronary angiography is increasingly being suggested. STD >0.1 mV in 8 or more surface leads associated with "left main coronary obstruction— lead aVR STE and lateral STD" is defined in the guidelines as "left main coronary obstruction— lead aVR STE and lateral STD."

CONCLUSION

Finally, the ECG provides a snapshot of the heart's electrical activation, which may be influenced by active ischemia. As a result, ECG interpretation provides only a hazy picture of myocardial ischemia's presence, severity, and extent The alterations are sometimes obvious and widespread. In other cases, the changes are modest and it can only be seen when reviewing an ECG recording after the severity of symptoms has changed (worsening or resolution of symptoms). ECG interpretation is a crucial component of the initial evaluation of patients with symptoms which are thought to be related to myocardial ischemia, along with a comprehensive history and physical examination (pain, shortness of breath, dizziness, etc.). The accuracy of reading is improved by the availability of earlier ECG tracing and repeated ECG.

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